

ENVS 193DS Homework 3

Danielle Sur

2026-05-31

Table of contents

Set up	2
Problem 1. Giant kelp fronds	2
a. An appropriate test	2
b. Create a visualization	2
c. Check your assumptions and run your test	2
d. Results communication	3
e. Test implications	3
f. Double check your own work	3
Problem 2. Personal data	3
a. Updating your visualizations	3
Cleaning data	3
Figure 1.	3
Figure 2.	5
b. Captions	6
Problem 3. Affective visualization	7
a. Describe in words what an affective visualization could look like for your personal data.	7
b. Create a sketch (on paper) of your idea.	7
c. Make a draft of your visualization.	7
d. Write an artist statement.	7
e. Prep your materials to share in class.	8
Problem 4. Statistical critique	8
a. Revisit and summarize	8
b. Visual clarity	9
c. Aesthetic clarity	9

Set up

```
# loaded in packages
library(tidyverse)
library(janitor)
library(here)
library(readxl)

# read in data and saved as object called `kelp`
kelp <- readr::read_csv(here::here("data","temp-kelp.csv"))
# read in my data and saved as object called 'rxntime'
rxntime <- readr::read_csv(here::here("data","rxn-time.csv"))
```

Problem 1. Giant kelp fronds

a. An appropriate test

Pearson's R and Pearson's product-moment correlation tests

ols

b. Create a visualization

S

c. Check your assumptions and run your test

a

d. Results communication

e. Test implications

f. Double check your own work

Problem 2. Personal data

a. Updating your visualizations

Cleaning data

```
# created a new object from rxntime
rxntime_type <- rxntime |>
  # cleaned column names
  clean_names() |>
  # selected columns of interest
  select(exercise_type, reaction_time_avg_ms) |>
  # removed missing values
  drop_na(exercise_type, reaction_time_avg_ms) |>
  # capitalized existing exercise type categories
  mutate(exercise_type = case_when(
    exercise_type == "biking" ~ "Biking",
    exercise_type == "volleyball" ~ "Volleyball",
    exercise_type == "walking" ~ "Walking"
  ))
```

```
# created a new object from rxntime
rxntime_clean <- rxntime |>
  # cleaned column names
  clean_names() |>
  # selected columns of interest
  select(exercise_duration_min, reaction_time_avg_ms)
```

Figure 1.

```

# base layer: ggplot
ggplot(data = rxn_time_type,
      # x-axis: exercise type
      # y-axis: average reaction time in milliseconds
      # color by exercise type
      mapping = aes(x = "Exercise Type",
                    y = "Average Reaction Time (ms)",
                    color = exercise_type)) +
# first layer: data points
geom_point(data = rxn_time_type,
          aes(x = exercise_type,
              y = reaction_time_avg_ms,
              size = 3)) +
# changed theme
theme_minimal() +
# removed legend
theme(legend.position = "none",
      panel.background = element_blank(), # removing panel background
      panel.grid = element_blank(), # removing panel grid
      axis.ticks = element_blank() # removing axis ticks)
) +
# relabeled x- and y- axes, adding title
labs(title = "Exercise Type and Reaction Time",
      subtitle = "May 29, 2026",
      x = "Exercise Type",
      y = "Average Reaction Time (ms)") +
# custom colors
scale_color_manual(values =
                  c("Biking" = "magenta",
                    "Volleyball" = "purple3",
                    "Walking" = "blue"))

```

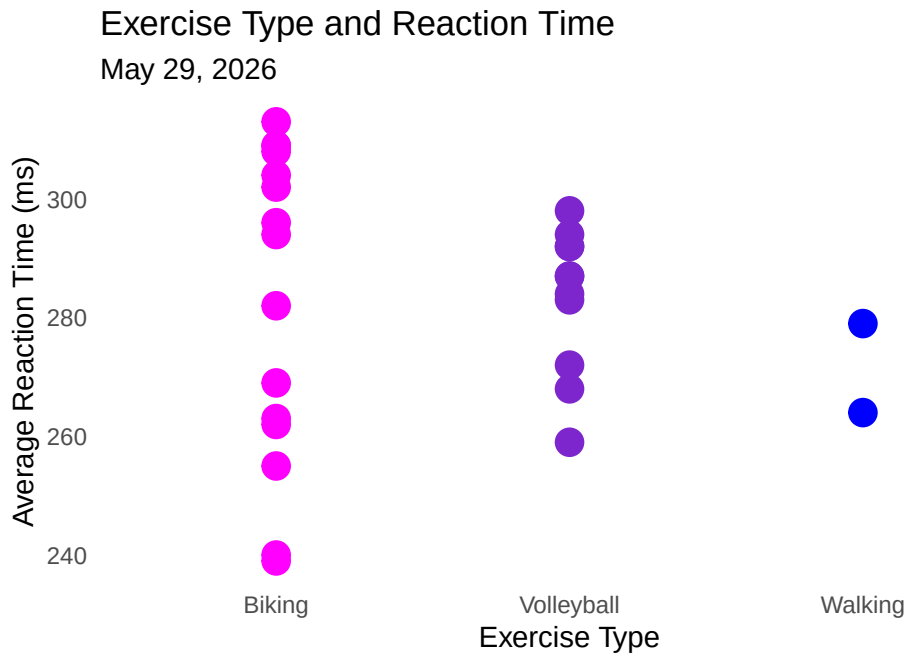


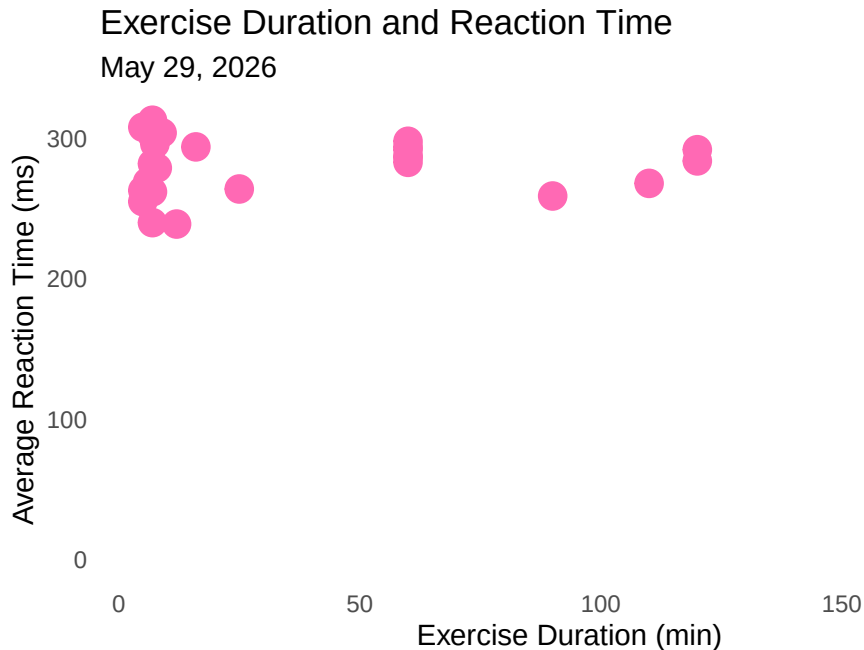
Figure 2.

```
# base layer: ggplot
ggplot(data = rxntime_clean,
  # x-axis: exercise duration in minutes
  # y-axis: average reaction time in milliseconds
  # color
  mapping = aes(x = "Exercise Type",
    y = "Average Reaction Time (ms)")) +
# first layer: data points
geom_point(data = rxntime_clean,
  aes(x = exercise_duration_min,
    y = reaction_time_avg_ms,
    size = 3,
    color = "hotpink")) +
# changed theme
theme_minimal() +
# removed legend
theme(legend.position = "none",
  panel.background = element_blank(), # removing panel background
  panel.grid = element_blank(), # removing panel grid
```

```

    axis.ticks = element_blank(), # removing axis ticks
  ) +
  # relabeled x- and y- axes, adding title
  labs(title = "Exercise Duration and Reaction Time",
        subtitle = "May 29, 2026",
        x = "Exercise Duration (min)",
        y = "Average Reaction Time (ms)") +
  scale_color_manual(values = "hotpink")

```



b. Captions

Figure 1. Mean reaction times across biking, volleyball, and walking exercises. Points represent observations of mean reaction time (ms) after different kinds of exercise (pink: Biking, purple: volleyball, blue: walking). Data from my personal observations. Data accessed 2026-05-29.

Figure 2. Mean reaction times and exercise duration. Points represent observations of mean reaction time (ms) in relation to exercise duration (min). Data from my personal observations. Data accessed 2026-05-29.

Problem 3. Affective visualization

a. Describe in words what an affective visualization could look like for your personal data.

Taking inspiration from the Dear Data project, I could make a visual where each observation is represented by an icon (a stick figure for walking, a volleyball, and a bike tire for biking). Additional doodles such as a water droplet or a star can indicate whether water was consumed throughout exercise, or if an exercise exceeded a certain amount of time. Colors can be used to indicate levels of social interaction.

b. Create a sketch (on paper) of your idea.

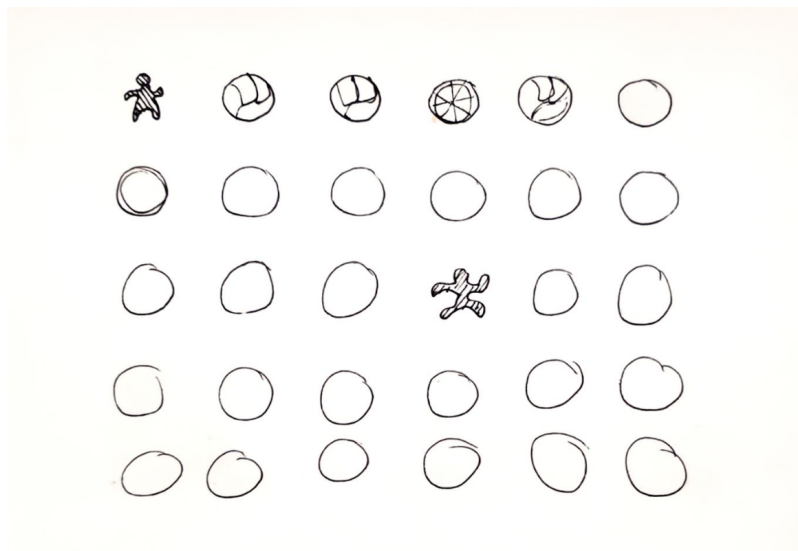


Figure 1

c. Make a draft of your visualization.

d. Write an artist statement.

This is a visualization of the exercises I have done throughout the spring quarter. The icons arranged in an array of thirty observations each represent a type of exercise (walking, biking, or volleyball) while indicating factors such as socialization with other people and hydration. I was inspired most by the Dear Data project; especially by how colors and shapes were used not only to embellish the visualization but to tell a story. I created this draft with a sheet of A4 paper, a pen, and a couple highlighters. It is hand-drawn.

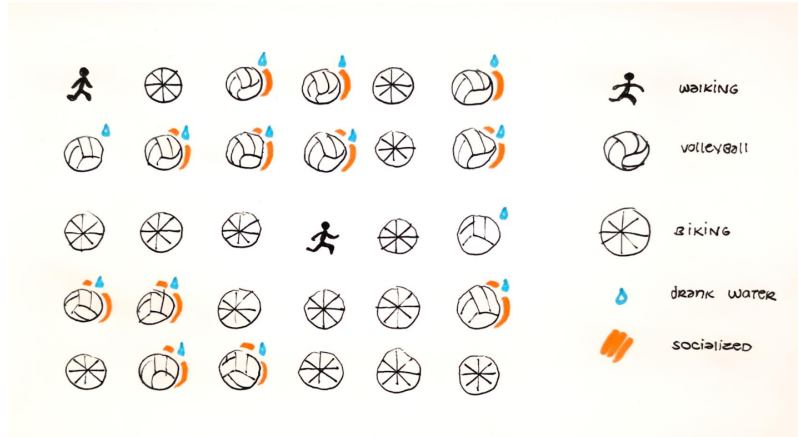


Figure 2

e. Prep your materials to share in class.

View presentation slides [here](#).

Problem 4. Statistical critique

a. Revisit and summarize

Pearson's R tests were used in this experiment to determine whether the relationship between coral cover and sea urchin density across all six studied reefs (Malindi, Watamu, Vipingo, Kanamai, Bamburi, and Diani) was significant. A significant result ($p < 0.05$) indicates a notable correlation. Using urchin density and other data (i.e. density of urchin predators and coral topographical complexity), these findings can be attributed to a complex interplay of biological factors.

Table 2. Water depth at low tide, topographic complexity (rugosity), the percent cover ($\bar{x} \pm \text{SD}$) of substrate categories and substrate diversity in the six locations. Sample sizes are given in Table 1. The significance level of a Kruskal-Wallis test of difference between reefs is provided

	Malindi	Watamu	Vipingo	Kanamai	Bamburi	Diani	Kruskal-Wallis
Depth, m	0.8 $\pm$ 0.3	1.1 $\pm$ 0.2	0.4 $\pm$ 0.2	0.4 $\pm$ 0.1	1.1 $\pm$ 0.3	0.6 $\pm$ 0.1	
Topographic complexity	1.37 $\pm$ 0.33	1.45 $\pm$ 0.20	1.30 $\pm$ 0.16	1.24 $\pm$ 0.18	1.23 $\pm$ 0.15	1.15 $\pm$ 0.12	$p < 0.001$
Hard coral	17.6 $\pm$ 12.8	30.0 $\pm$ 24.7	18.1 $\pm$ 13.6	12.7 $\pm$ 9.5	8.9 $\pm$ 6.9	4.6 $\pm$ 4.6	$p < 0.001$
Algal turf	27.6 $\pm$ 20.5	19.7 $\pm$ 17.5	58.6 $\pm$ 19.8	44.3 $\pm$ 15.4	66.3 $\pm$ 14.8	54.7 $\pm$ 22.2	$p < 0.001$
Calcareous algae	8.3 $\pm$ 6.3	10.3 $\pm$ 10.3	0.6 $\pm$ 1.5	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	1.4 $\pm$ 3.1	$p < 0.001$
Macro algae	2.4 $\pm$ 3.2	2.5 $\pm$ 2.9	2.8 $\pm$ 4.2	0.9 $\pm$ 2.4	2.4 $\pm$ 3.0	1.5 $\pm$ 1.9	$p < 0.05$
Seagrass	18.1 $\pm$ 22.6	28.1 $\pm$ 32.9	13.8 $\pm$ 14.6	31.1 $\pm$ 18.9	9.2 $\pm$ 15.6	31.9 $\pm$ 23.0	$p < 0.001$
Coralline algae	11.6 $\pm$ 14.0	0.2 $\pm$ 0.7	0.04 $\pm$ 0.2	0.1 $\pm$ 0.4	2.2 $\pm$ 2.8	0.3 $\pm$ 0.7	$p < 0.001$
Soft coral	1.4 $\pm$ 2.4	0.0 $\pm$ 0.0	1.9 $\pm$ 3.0	1.8 $\pm$ 3.6	3.0 $\pm$ 3.8	0.8 $\pm$ 1.2	$p < 0.006$
Coral sand	14.6 $\pm$ 17.0	9.1 $\pm$ 9.6	4.3 $\pm$ 7.3	10.1 $\pm$ 10.1	8.2 $\pm$ 10.9	4.7 $\pm$ 8.4	$p < 0.02$
Sponge	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.6 $\pm$ 1.1	0.1 $\pm$ 0.4	0.5 $\pm$ 1.1	0.04 $\pm$ 0.2	$p < 0.008$
Substrate diversity	0.82 $\pm$ 0.0	0.78	0.59 $\pm$ 0.1	0.67 $\pm$ 0.0	0.53 $\pm$ 0.1	0.60 $\pm$ 0.0	$p = 0.07$

Figure 3

b. Visual clarity

c. Aesthetic clarity

d. Recommendations